

In-Class Exercise: Hydraulic Bottle Jack

Mechanical and Hydraulic Systems

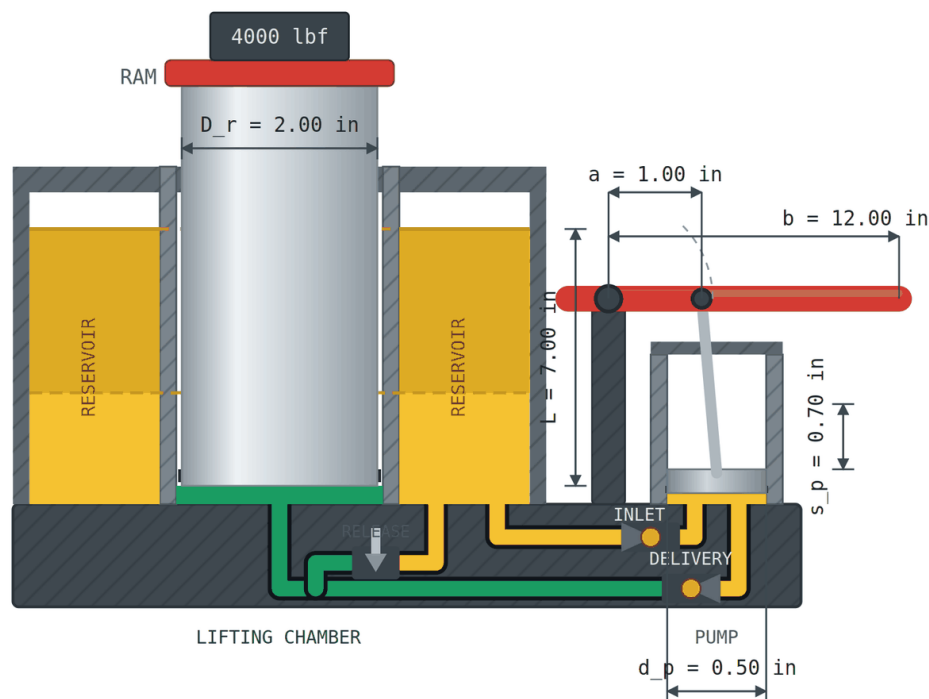
Name _____ Date _____

A 2 ton bottle jack has been measured and the dimensions are marked on the cross section below. Work the calculations by hand first, in the order given. Then open the simulation, enter the same dimensions in the panel on the right, and compare each of your answers with the Laboratory calculations panel. Every value you need is on the figure or in the list under it.



<https://alibulentkoc.github.io/hydraulic-bottle-jack/simulator/hydraulic-bottle-jack.html>

Scan or type. Runs in any browser, no installation.



Amber is reservoir oil at atmospheric pressure. Green is oil at lifting pressure. The boundary between the two colors is the delivery check ball.

Given

1. Rated load $W = 4000$ lbf, printed on the jack
2. Lifting cylinder bore $D_r = 2.00$ in
3. Pump cylinder bore $d_p = 0.50$ in
4. Pump piston stroke $s_p = 0.70$ in
5. Lifting cylinder extension $L = 7.00$ in
6. Pivot to pump rod end $a = 1.00$ in, pivot to operator hand $b = 12.00$ in
7. Handle swept angle $\theta = 44.4$ deg, measured with an angle gauge
8. Reservoir annulus, not shown: inside diameter $D_i = 2.10$ in, outside diameter $D_o = 3.50$ in, oil column height $H = 6.00$ in
9. Pumping rate $n = 1$ full cycle per second. Use $1 \text{ hp} = 6600 \text{ in-lb/s}$.

Calculate

1. Pump piston area A_p and lifting cylinder area A_r .
2. Volume displaced per full pump stroke, $V_p = A_p s_p$.
3. Lift produced per stroke, $h = V_p / A_r$. State which principle this uses.

4. Pressure in the lifting cylinder at the rated load, $P = W / A_r$.
5. Force on the pump piston, $F_p = P A_p$. State which principle lets you use the same pressure on both pistons.
6. Torque at the handle pivot, $T = F_p a$, and the force at the operator's hand, $F_{hand} = T / b$.
7. Number of strokes and the time to reach full extension, then the speed of extension.
8. Hydraulic power P_Q , output linear power W_v , and operator power from the arc the hand travels, $F_{hand} (2 \pi b \theta / 360) n$. Convert each to horsepower.
9. Reservoir annulus area, the drop in oil level at full extension, and whether the reservoir holds enough oil for one full extension.
10. Force multiplication W / F_{hand} . Show that it equals the area ratio times the lever ratio.

Check in the simulation

Switch the units to Customary and enter the eleven values from the list above. The Laboratory calculations panel then works the same quantities, in nearly the same order. Mark any answer that differs from yours and find the reason before moving on.

Explore

1. Predict, then change the lifting cylinder bore to 4.00 in. What happens to the lift per stroke, the number of strokes, and the hand force? Explain all three from one principle.
2. Restore 2.00 in, then halve the pump bore to 0.25 in. Predict the hand force before you look.
3. Your three power figures should be one number for an ideal machine. Which of the three differs, by what factor, and why? The panel shows the factor.